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ECEN 3730 – PCB Design Manufacture  
11/10/25

## **Board 3 Report**

Objective:

The objective of designing Board 3, the *Golden Arduino*, is to complete a fully working custom Arduino-compatible PCB prototype. It should be able to receive uploaded code over its micro-USB port, utilize the Arduino IDE as normal, and be compatible with standard Arduino Uno R3 shields. This design required extensive reading and interpretation of datasheets, making independent design decisions, and implementing standard good engineering practices like including test points, LED indicators, and isolation switches. The board's Atmega328 processor must function correctly alongside a USB serial interface so that a bootloader can be successfully burned from an original programmer Arduino. This will allow the board to operate like its commercial Arduino Uno counterpart. Power regulation should be included as well as a functioning reset switch. After soldering, the design is validated by burning the bootloader, uploading sketches to verify functionality, and confirming correct operation and switching behavior by comparing results to an official Arduino.

Project Overview:

1. Design an Arduino compatible PCB
2. The ATmega328 processor needs to function
3. The CH340 chip should be implemented for USB compatibility
4. The TX/RX lines should be routed correctly (**TX → RX** NOT TX → TX)
5. The reset switch should be wired correctly
6. Test points and isolation switches should be included for good debugging practices
7. The 10 pin and 8 pin headers should have a 0.160" gap between them while the 8 pin and 6 pin headers should have a 0.200" gap between them. There should also be a 1.900" gap between the two sets to ensure compatibility with Arduino shields
8. The new ATmega328 processor has to be burned with the bootloader
9. Test sketches should be uploaded to ensure board functionality
10. All components must be soldered neatly and correctly as good practice

## Schematic and PCB:

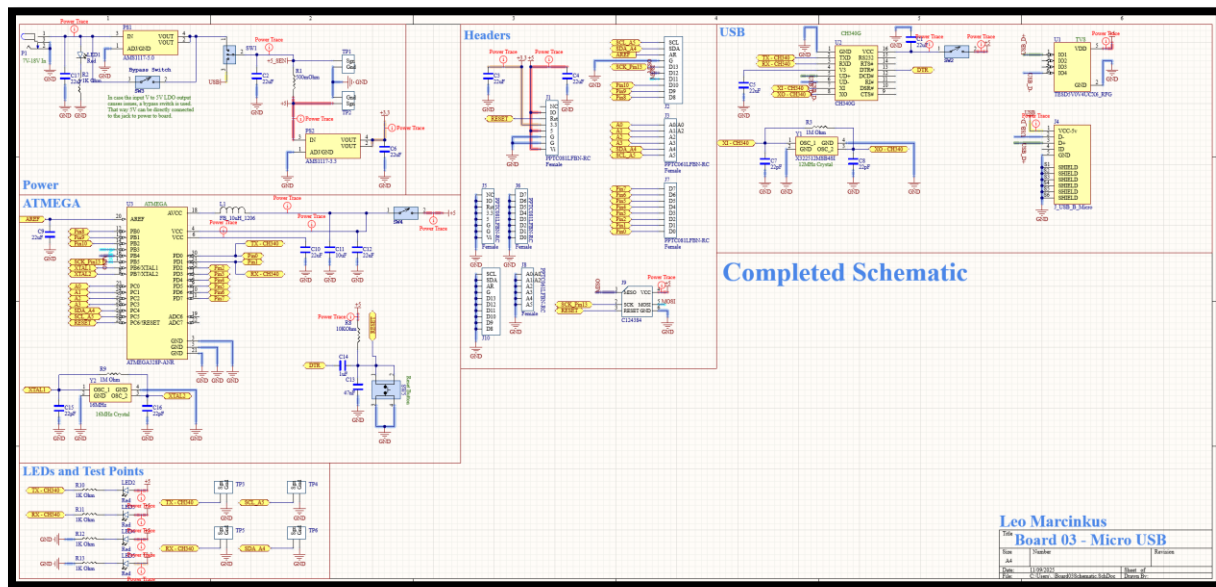


Figure 1: Altium Schematic

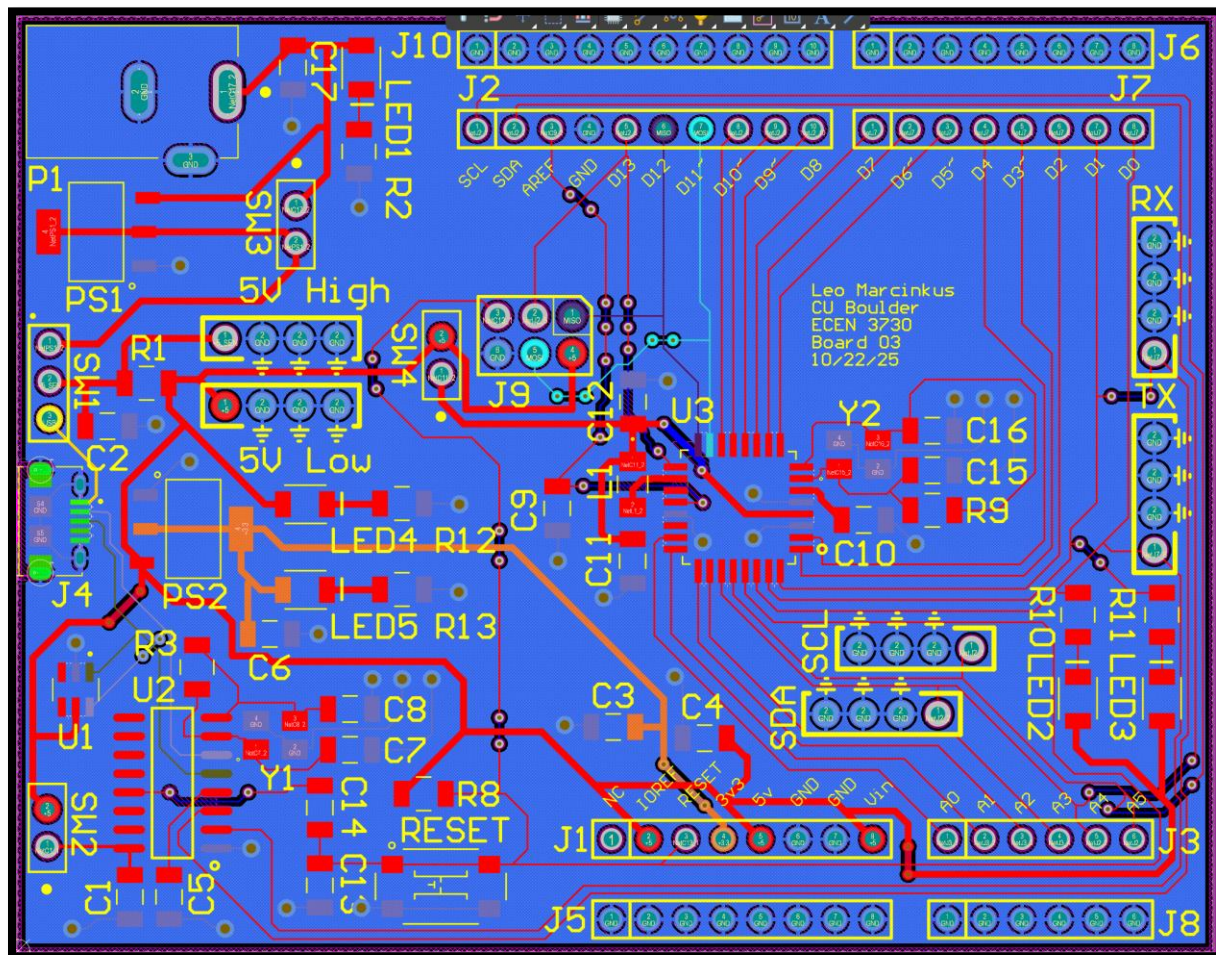


Figure 2: Altium PCB

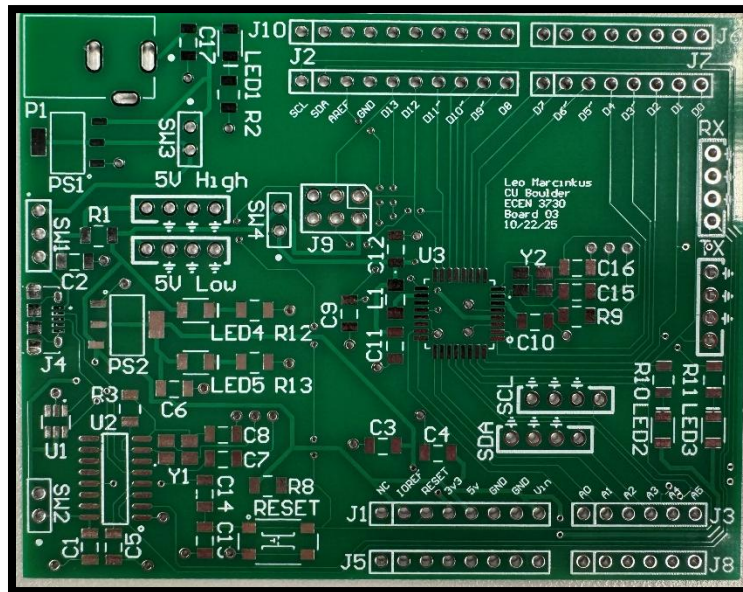


Figure 3: Unsoldered Board

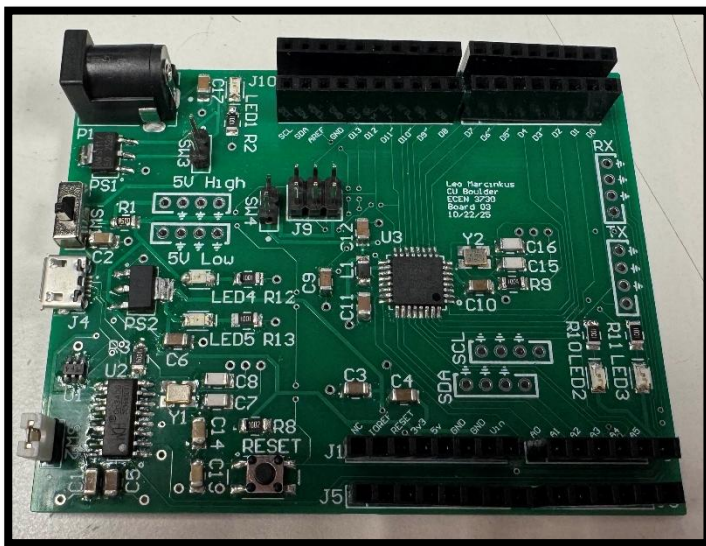


Figure 4: Soldered Board

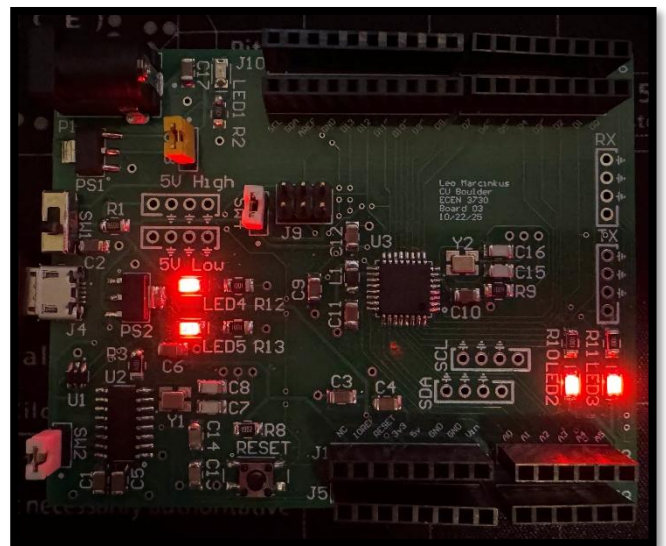


Figure 5: Powered Board

## Oscilloscope Measurements:

### Switching Noise on Quiet Low and Current Draw

#### Custom Board

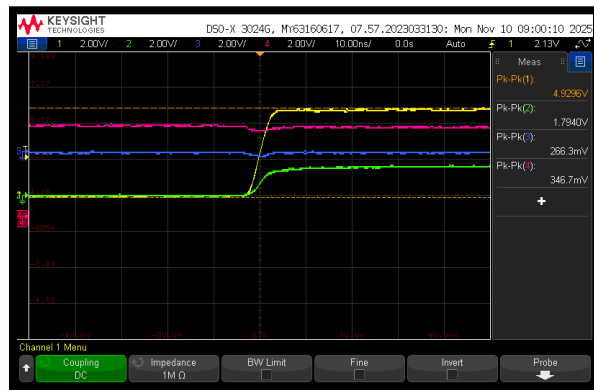


Figure 6: Rise → Ch1: Pin 13, Ch2:49 Ω, Ch3: Q-Low Ch4: Q-High

#### Official Board

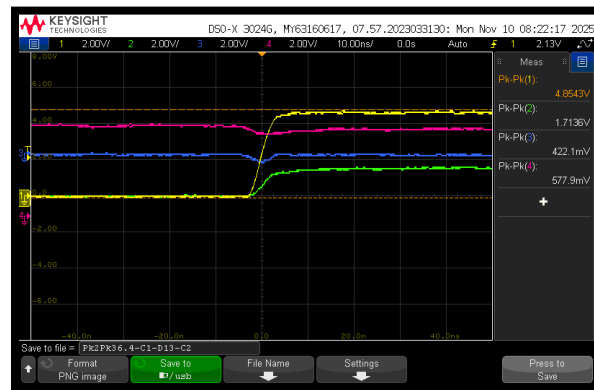


Figure 7: Rise → Ch1: Pin 13, Ch2:49 Ω, Ch3: Q-Low Ch4: Q-High

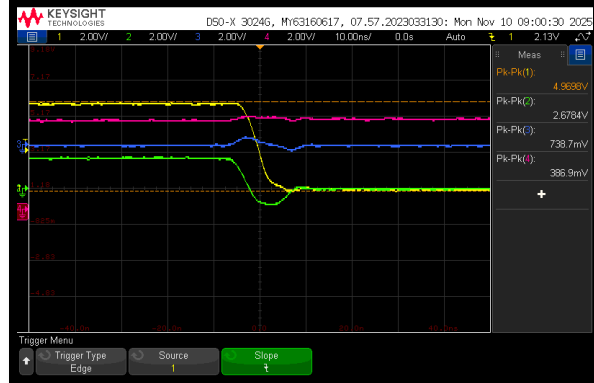


Figure 8: Fall → Ch1: Pin 13, Ch2:49 Ω, Ch3: Q-Low Ch4: Q-High

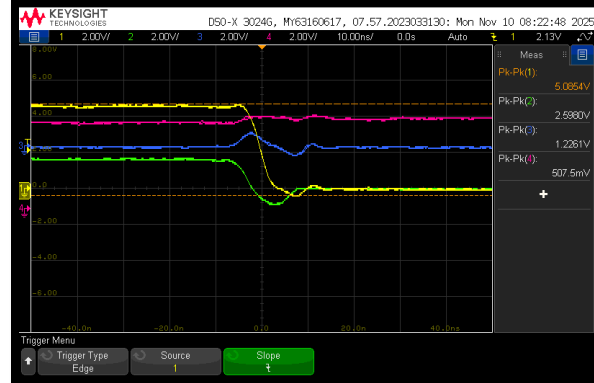


Figure 9: Fall → Ch1: Pin 13, Ch2:49 Ω, Ch3: Q-Low Ch4: Q-High

## Results:

The switching noise on the rising and falling edge of each board's Quiet Low and Quiet High is measured. As well as the voltage across a 49 Ω resistor to calculate its current. Pin 13 is used for the trigger.

| Custom Board       | Rising Edge | Falling Edge |
|--------------------|-------------|--------------|
| Pin 13 (Pk-Pk)     | 4.9296 V    | 4.9698 V     |
| 49 Ω (Pk-Pk)       | 1.7940 V    | 2.6784 V     |
| Quiet Low (Pk-Pk)  | 0.2663 V    | 0.7387 V     |
| Quiet High (Pk-Pk) | 0.3467 V    | 0.3869 V     |

| Official Board     | Rising Edge | Falling Edge |
|--------------------|-------------|--------------|
| Pin 13 (Pk-Pk)     | 4.8543 V    | 5.0854 V     |
| 49 Ω (Pk-Pk)       | 1.7136 V    | 2.5980 V     |
| Quiet Low (Pk-Pk)  | 0.4221 V    | 1.2261 V     |
| Quiet High (Pk-Pk) | 0.5779 V    | 0.5075 V     |

The current across the  $49\ \Omega$  resistor is found using Ohm's Law.

$$V = \frac{I}{R} \rightarrow I = VR$$

| Board:         | Rising Edge:                                      | Falling Edge:                                      |
|----------------|---------------------------------------------------|----------------------------------------------------|
| Custom Board   | $I_{Rising} = 1.7940\ V / 49\ \Omega = 0.0366\ A$ | $I_{Falling} = 2.6784\ V / 49\ \Omega = 0.0547\ A$ |
| Official Board | $I_{Rising} = 1.7136\ V / 49\ \Omega = 0.0350\ A$ | $I_{Falling} = 2.5980\ V / 49\ \Omega = 0.0530\ A$ |

Using these results, the ground-bounce per amp can be calculated. This will determine which board performed better by finding out which has the least amount of noise given the same load. Since the equation calculates the ratio of noise per amp, smaller is better.

$$Ratio = \frac{Quiet\ Low_{Switching\ Noise}}{I_{Peak\ through\ 49\ \Omega}}$$

| Board/Edge:    | Rising Edge:                              | Falling Edge:                             |
|----------------|-------------------------------------------|-------------------------------------------|
| Custom Board   | $Ratio = 0.2663\ V / 0.0366\ A = 7.2760$  | $Ratio = 0.7387\ V / 0.0547\ A = 13.5046$ |
| Official Board | $Ratio = 0.4221\ V / 0.0350\ A = 12.0600$ | $Ratio = 1.2261\ V / 0.0530\ A = 23.1340$ |

It is clear that the custom board outperforms the official board on both the rising and falling edge. It performs  $\sim 1.6575x$  better than the official board on the rising edge and  $\sim 1.7130x$  better than the official board on the falling edge.

## Noise on the Power Rail when the Microcontroller is the Aggressor

### Custom Board



Figure 10: Rise → Ch1: Pin 13, Ch2: Q-High Ω, Ch4: 5V M1: Ch2 - Ch4

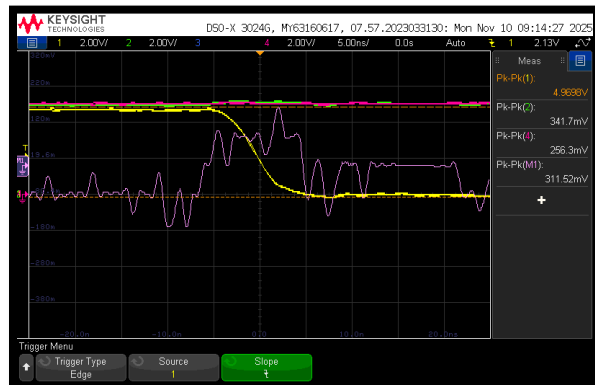


Figure 12: Fall → Ch1: Pin 13, Ch2: Q-High Ω, Ch4: 5V M1: Ch2 - Ch4

### Official Board



Figure 11: Rise → Ch1: Pin 13, Ch2: Q-High Ω, Ch4: 5V M1: Ch2 - Ch4



Figure 13: Fall → Ch1: Pin 13, Ch2: Q-High Ω, Ch4: 5V M1: Ch2 - Ch4

### Results:

The chip and its I/O are the aggressor source of noise on the power rail. The voltage on the Quiet High pin and the voltage on the board power rail is compared. I did not offset the Quiet High and 5V to show how similar they are. The M1 line is Quiet High – 5V, representing the voltage difference showing how much noise is dropped across the PDN and die-to-board return path.

| Custom Board         | Rising Edge | Falling Edge |
|----------------------|-------------|--------------|
| Pin 13 (Pk-Pk)       | 4.9296 V    | 4.9698 V     |
| Quiet High (Pk-Pk)   | 0.3317 V    | 0.3417 V     |
| 5V (Pk-Pk)           | 0.1910 V    | 0.2563 V     |
| M1 (Quiet High – 5V) | 0.46738 V   | 0.31152 V    |

| Official Board       | Rising Edge | Falling Edge |
|----------------------|-------------|--------------|
| Pin 13 (Pk-Pk)       | 4.8141 V    | 5.0503 V     |
| Quiet High (Pk-Pk)   | 0.6231 V    | 0.5427 V     |
| 5V (Pk-Pk)           | 0.1508 V    | 0.1809 V     |
| M1 (Quiet High – 5V) | 0.84922 V   | 0.77891 V    |

The Quiet High is measuring the on-die VDD rail, inside the ATmega328. The 5V is measurement is measuring the board-level VDD rail on the noise shield being used. Comparing the Quiet High measurements, it is clear that the custom board performs better. On the rising edge, the custom board performs  $\sim 1.8785x$  better than the official board. On the falling edge, the custom board performs  $\sim 1.5882x$  better than the official board. Meaning that the custom board has essentially **half** of the on-die noise in the ATmega328 chip than the official board.

However, when measuring the board-level 5V rail noise, the official board performs slightly better. On the rising edge, the official board performs  $\sim 0.7895x$  better than the custom board. On the falling edge, the official board performs  $\sim 0.7058x$  better than the custom board. Meaning that the official board has less overall disturbance affecting the 5V rail.

Although the custom board may produce less overall, it allows more noise to escape the chip. Despite the custom board's advantage, it means its chip is less isolated and leaks more noise onto its 5V rail. If the goal is to protect the chip, then the custom board performs better, but if the focus is keeping a clean power rail, then the official board performs better.

### Noise on the Power Rail when the Board is the Aggressor

#### Custom Board

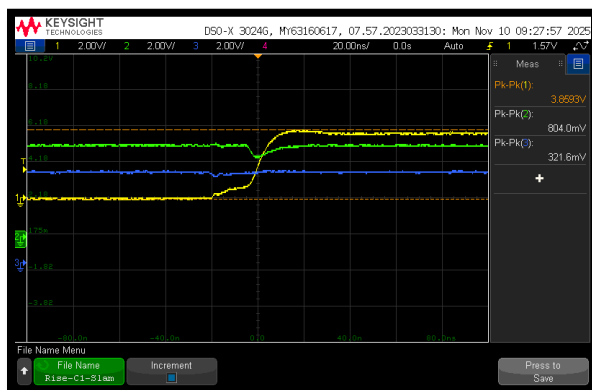


Figure 14: Rise  $\rightarrow$  Ch1: Slam, Ch2: 5V  $\Omega$ , Ch3: Q-High

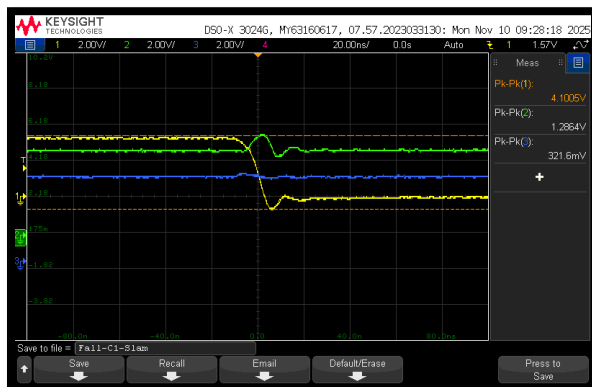


Figure 16: Fall  $\rightarrow$  Ch1: Slam, Ch2: 5V  $\Omega$ , Ch3: Q-High

#### Official Board

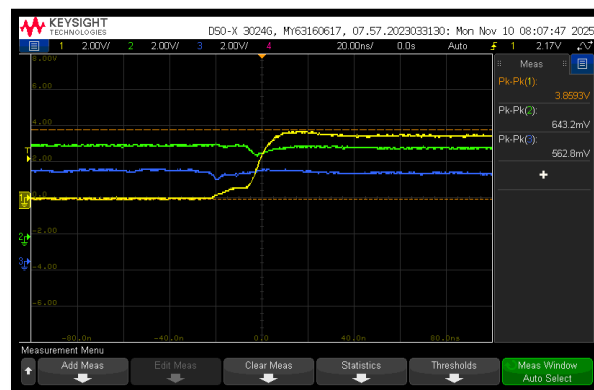


Figure 15: Rise  $\rightarrow$  Ch1: Slam, Ch2: 5V  $\Omega$ , Ch3: Q-High

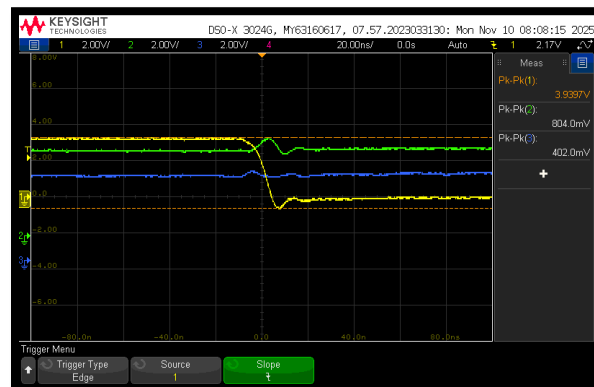


Figure 17: Fall  $\rightarrow$  Ch1: Slam, Ch2: 5V  $\Omega$ , Ch3: Q-High

## Results:

Digital pin D7 connects to a MOSFET. The MOSFET is going to draw current on the 5V rail from the board. When D7 turns on, the MOSFET turns on and current flows through a 10  $\Omega$  sense resistor. This time, the noise is being created on the board's power rail so the 5V rail on the board is the aggressor. The noise on the 5V rail on the board and how much is on the die is measured. The scope is triggered (Ch 1) with voltage going across a 10  $\Omega$  sense resistor.

| Custom Board       | Rising Edge | Falling Edge |
|--------------------|-------------|--------------|
| Slam (Pk-Pk)       | 3.8593 V    | 4.1005 V     |
| 5V (Pk-Pk)         | 0.8040 V    | 1.2864 V     |
| Quiet High (Pk-Pk) | 0.3216 V    | 0.3216 V     |

| Official Board     | Rising Edge | Falling Edge |
|--------------------|-------------|--------------|
| Slam (Pk-Pk)       | 3.8593 V    | 3.9397 V     |
| 5V (Pk-Pk)         | 0.6432 V    | 0.8040 V     |
| Quiet High (Pk-Pk) | 0.5628 V    | 0.4020V      |

Using these results, the coupling ratio can be calculated (smaller is better).

$$Ratio = \frac{Quiet\ High}{V_{5V\ Droop}}$$

| Board/Edge:    | Rising Edge:                             | Falling Edge:                            |
|----------------|------------------------------------------|------------------------------------------|
| Custom Board   | $Ratio = 0.3216\ V / 0.8040\ V = 0.4000$ | $Ratio = 0.3216\ V / 1.2864\ V = 0.2500$ |
| Official Board | $Ratio = 0.5628\ V / 0.6432\ V = 0.8750$ | $Ratio = 0.4020\ V / 0.8040\ V = 0.5000$ |

The custom board outperforms the official board on both the rising and falling edge. It performs ~2.1875x better than the official board on the rising edge and 2.0x better than the official board on the falling edge. However, these results are only great if the goal is to protect the microcontroller's internal rail. If the desired outcome is having a board with a strong PDN then the official board outperforms the custom board since the official board's 5V droop is significantly less on both edges when compared.

## Near Field Emissions from the Bottom of each Board

**Custom Board**

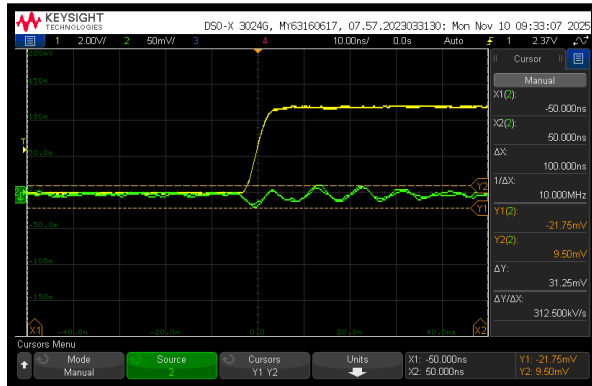


Figure 18: Rise → Ch1: Pin 13, Ch2: Shorted GND

**Official Board**

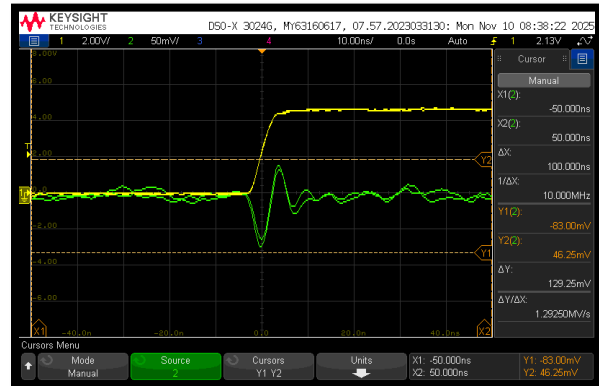


Figure 19: Rise → Ch1: Pin 13, Ch2: Shorted GND

### Results:

The final measurement utilizes a 10x probe shorted to its own ground to create a small loop. This acts as a small pick-up probe to measure the near field emissions from the board. With the code running and Pin 13 triggering the scope, the pick-up probe is aligned underneath the board to generate noise. It is placed in a spot where the most noise can be measured.

|                          |             |
|--------------------------|-------------|
| Custom Board             | Rising Edge |
| RF Pick-up Noise (Pk-Pk) | 0.03125 V   |

|                          |             |
|--------------------------|-------------|
| Official Board           | Rising Edge |
| RF Pick-up Noise (Pk-Pk) | 0.12925 V   |

The custom board performs 4.1360x better than the official board. This aligns with earlier results of the custom board producing less rail bounce preventing further excitation when a small magnetic-field is introduced due to channel 2's loop.

### Measurement Analysis:

There is a noticeable difference in performance between the two boards.

### Switching Noise on Quiet Low and Current Draw:

- The custom board performs  $\sim 1.6575x$  better than the official board on the rising edge
- The custom board performs  $\sim 1.7130x$  better than the official board on the falling edge

### Noise on the Power Rail when the Microcontroller is the Aggressor:

- The custom board performs  $\sim 1.8785x$  better than the official board on the rising edge
- The custom board performs  $\sim 1.5882x$  better than the official board on the falling edge
- When measuring the board-level 5V rail noise, the official board performs  $\sim 0.7895x$  better than the custom board on the rising edge
- When measuring the board-level 5V rail noise, the official board performs  $\sim 0.7058x$  better than the custom board on the falling edge
- Depends on the intended priority of the board. If the goal is to protect the chip, then the custom board performs better, but if the focus is keeping a clean power rail, then the official board performs better

### Noise on the Power Rail when the Board is the Aggressor:

- The custom board performs  $\sim 2.1875x$  better than the official board on the rising edge
- The custom board performs  $\sim 2.0x$  better than the official board on the falling edge
- The official board has significantly less 5V droop than the custom board
- Depends on the intended priority of the board. If the goal is to have a board with a strong PDN then the official board outperforms the custom board

### Near Field Emissions:

- The custom board performs  $4.1360x$  better than the official board

I suspect the performance difference to be in favor of the custom board in the first experiment due to better local decoupling compared to the official board. When the microcontroller is the aggressor, it is clear that the official board has a cleaner power rail due to a stronger PDN. When the board is the aggressor, the custom board exhibits its better high-frequency isolation. Most likely due to shorter current paths compared to the official board. Similar for the near field emissions measurement, I believe the custom board outperformed the official board due to tighter return paths, decoupling capacitor placements, and more efficient via stitching.

#### What Worked:

1. The board entirely functions
2. The ATmega328 chip was successfully burned
3. The board is recognized in the Arduino IDE and code can be uploaded and run
4. The board is compatible with Arduino Uno R3 shields

#### Analysis of Project:

1. *What worked well and should be carried over to the next board?*
  - a. Clear labels made assembly of the board easy as it was simple to tell what component should go where. Separating sections of the board significantly organized the design of the circuit. Including multiple indicator LEDs and isolation switches made it clear what parts of the circuit were functioning correctly.
2. *What should be done differently?*
  - a. Although there were no issues, I think better component placement during the design phase would be beneficial. I tried to make a more compact board, but it made it significantly harder to assemble. With less space between components, I should have shrunk the size of component labels, so a few are not so far from their corresponding component. Better trace layout should be implemented to use fewer vias.

3. *Any hard errors?*

- a. No hard errors occurred preventing function.

4. *Any soft errors that can be corrected for the next board?*

- a. No specific errors occurred when recording data. Using net labels instead of ports on the next board will be beneficial. It will make the PCB design quicker in Altium since a few lines were not labeled. It was tedious to keep referring back to the schematic.

Conclusion:

The custom “Golden Arduino” Board 3 was successfully designed and assembled. It correctly meets each function and design requirement allowing the board to act as a stand-alone Arduino. After successfully burning the bootloader, functionality testing sketches were run to ensure correct operation. Multiple experiments were conducted and results were compared to an official board. The measured data shows key improvements in several aspects compared to its official counterpart, most notably switching noise and near-field emissions. This increase in performance is likely due to design changes such as tighter return paths and better local decoupling. The official board, however, still demonstrated a stronger PDN and less power rail distortion. Overall, this project heavily expanded my design, debugging, and testing skills. It allowed me to investigate the performance differences that seemingly small design changes can instill.